Exact and Bounded Stability Audits for Pairwise Leaderboards

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Abstract

A small number of changed comparisons can reverse a leaderboard, but the meaning of "small" depends on score separation, exposure, and the perturbation model. We audit fixed-schedule outcome reversals in ridge Bradley–Terry rankings across crowdsourced preferences, expert judgments, model judgments, and basketball results. A direct-transfer relaxation gives a finite-edit lower bound: no set of k outcome reversals reduces a queried score gap more than k unconstrained transfers between its endpoints. When enough appropriate direct comparisons exist, the bound is attainable and the minimum reversal budget is exact. Across nine main panels, we certify the exact winner-change budget in 6. These budgets include 2 records for the Arena snapshot and 46 and 61 for expert and GPT-4 judgments. Arena's leading gap is already unresolved under resampling; its two-edit threshold becomes 27 for a reversal beyond twice its original bootstrap standard deviation. Matching judgment schedules preserves a larger GPT-4 threshold in these data. We find substantial variation within designs, and show why global edit percentages, participant exposure, and sampling uncertainty must be reported together.

1 Introduction

Pairwise leaderboards turn comparisons into consequential ordinal statements: which model to deploy, which release is ahead, or which team is strongest. In preference evaluation, the observations are judgments rather than ground truth. Chatbot Arena gathers anonymous user comparisons and estimates model strengths from them [Chiang et al., 2024]; MT-Bench offers expert and model-generated judgments on a controlled question set [Zheng et al., 2023]. The same mathematical ranking model also applies to sporting outcomes. An ordering can change because a close contest is intrinsically uncertain, because a participant has little supporting data, or because particular observations have disproportionate influence. These mechanisms should not be conflated.

Our question is operational: *with the observed comparison schedule fixed, how many decisive outcomes must be reversed to change an estimated order, and how does that budget compare with random noise, participant exposure, and sampling uncertainty?* We study offline data sensitivity. The ability to rewrite arbitrary historical records is stronger than a practical voter's access. A reversal budget is consequently an audit of an estimator and dataset, not an estimate of the effort required to compromise a live service.

We distinguish the reversal of the original leading pair from the loss of the original winner to *any* competitor. Searching for a successful edit set gives an upper bound on the optimal budget, not evidence that fewer edits are impossible. We derive a complementary lower bound for the standard ridge Bradley–Terry model. Its Hessian is a comparison-graph Laplacian plus a positive diagonal. A maximum principle implies that a direct transfer between the queried participants is the most effective unit transfer at every parameter value. Integrating this fact gives a global comparison with a virtual,

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

capacity-free direct-transfer path. Sufficient recorded direct wins make the relaxation attainable; otherwise we retain a lower/upper interval.

The empirical study uses a fixed Arena snapshot, both MT-Bench judgment panels, and the last six complete calendar-labeled NBA seasons in the supplied historical archive. It also uses a matched MT-Bench schedule, multiple perturbation baselines, cluster resampling, and modeling controls. We obtain exact or bounded answers rather than presenting a heuristic attack count as a universal robustness metric.

Three findings guide interpretation. First, the Arena leading pair is exceptionally close: its gap is 0.008, with a bootstrap interval $[-0.097, 0.116]$. A tiny point-order reversal is therefore a different event from overturning a well-separated result. Second, the denominator matters. NBA teams contribute a much smaller share of their board's comparisons than models do in the six-model MT-Bench panels; counting all records makes those differences look larger. Third, matching the human and GPT-4 schedules preserves the observed difference in leading-pair robustness, but cannot establish that a judge type is inherently better. Stability can also preserve systematic preferences or bias.

2 Related work

Chiang et al. [2024] describe preference collection, Bradley–Terry estimation, uncertainty, and sampling design in Chatbot Arena. Zheng et al. [2023] compare expert and LLM judgments, including order inconsistency and self-enhancement bias. These studies motivate comparing evaluation designs while avoiding the assumption that a more decisive judge is a more valid one.

Our candidate-selection baseline follows the general influence-function approach of Koh and Liang [2017]: differentiate a fitted optimum through its estimating equations. Broderick et al. [2023] use influence approximations to find influential deletions and emphasize exact refitting and the distinction between sampling uncertainty and sensitivity to selected subsets. We use this distinction directly. Our label-reversal threat leaves the comparison graph unchanged, whereas deletion changes both its weights and its outcomes. We report both operations but do not equate their budgets.

Min et al. [2025] study new-vote manipulation in Arena, including effects through comparisons that do not involve the target. Our contribution is not the observation that pairwise rankings can be manipulated. We analyze a fixed-schedule historical-edit model, derive a finite-edit lower bound with an attainability test, and compare its consequences across real evaluation designs with exposure and uncertainty controls. The theorem applies to item-only ridge Bradley–Terry fits; it does not claim optimality for new-vote collection, arbitrary covariates, or deletions. To preserve the requested independent research boundary, literature access was confined to selected supplied starting references and dataset documentation; this is not an exhaustive literature review.

3 Audit model and finite-edit bounds

3.1 Estimator, edits, and endpoints

Record r compares items a_r, b_r, with $x_r = e_{a_r} - e_{b_r}$ and outcome $y_r \in \{0, 1/2, 1\}$. A tie contributes one half-win observation, with the same total loss weight as a decisive record. The fitted score vector is

$$\hat{\theta} = \arg\min_{\theta \in \mathbb{R}^M} F(\theta), \quad F(\theta) = \sum_{r=1}^{N} \left[\log(1 + e^{x_r^\top \theta}) - y_r x_r^\top \theta \right] + \frac{\lambda}{2} \|\theta\|_2^2. \tag{1}$$

We use natural-logit scores, $\lambda = 1$ on the *summed* loss, and controls at 0.1 and 10. Positive ridge makes every fit unique even after unfavorable edits; it also affects very close rankings. Scores sum to zero automatically. These are our standardized reconstructed boards, not replicas of any current platform's production leaderboard.

An eligible edit reverses one originally decisive record, $y_r \mapsto 1 - y_r$, once. It preserves participants, schedule, and N; ties are not editable. For an edit set E,

$$F_E(\theta) = F(\theta) + b_E^\top \theta, \qquad b_E = \sum_{r \in E} (2y_r - 1) x_r = \sum_{r \in E} (e_{w_r} - e_{l_r}). \tag{2}$$

Here w_r, l_r are the original winner and loser. Fix the original leader ℓ and runner-up r_0, and put $c = e_\ell - e_{r_0}$ and $\hat{\Delta} = c^\top \hat{\theta} > 0$. Our primary pair endpoint is a strict reversal, $c^\top \hat{\theta}_E < 0$. Let $K^*_{\ell r_0}$ be its minimum feasible edit count. Separately, $K^*_{\text{win}} = \min_{j \neq \ell} K^*_{\ell j}$ measures loss to any competitor. Exact score ties do not count as reversals.

We report k, k/N, and k/n_ℓ, where n_ℓ is the number of original records involving the leader, including ties. The last is an exposure scale, not a probability or a claim that all edits must touch the leader. For the reported leading-pair static paths, all selected edits do touch it. None of these denominators equates human effort, corruption probability, or information across domains.

3.2 A direct-transfer relaxation

For $t \geq 0$ define the virtual fit

$$\theta_c(t) = \arg\min_\theta \{F(\theta) + t c^\top \theta\}, \quad L_c = \min\{k \in \mathbb{N} : c^\top \theta_c(k) < 0\}. \tag{3}$$

The virtual path can transfer more wins than were observed, even when the pair never met. It is a mathematical relaxation, not generated evidence. Strong convexity makes these virtual fits well-defined.

Theorem 1 (Direct-transfer domination)**.** *Under Eq.* (1) *with $\lambda > 0$, item effects only, and the fixed-schedule reversal model in Eq.* (2)*, for every set E of k eligible edits and any contrast $c = e_i - e_j$,*

$$c^\top \hat{\theta}_E \;\geq\; c^\top \theta_c(k). \tag{4}$$

Consequently $K^*_{ij} \geq L_c$. If at least L_c recorded decisive outcomes are wins by i against j, then $K^*_{ij} = L_c$.

Proof sketch. Write $H(\theta) = L(\theta) + \lambda I$ where L is a graph Laplacian with nonnegative logistic curvature weights. The solution $v = H(\theta)^{-1} c$ has its maximum at i and its minimum at j. Hence $v^\top (e_w - e_l) \leq v^\top c$ for every possible outcome direction. Interpolate the linear perturbation from b_E to kc and differentiate its optimum. The queried gap cannot increase along that interpolation, proving Eq. (4). The gap on the direct path decreases strictly because its derivative is $-c^\top H^{-1} c < 0$. If enough direct wins exist, reversing them attains that path. Appendix A supplies the full proof.

The result controls all allowable edit sets, not just a locally sorted path. We find L_c by bracketing and integer bisection with exact refits. Recorded direct-win capacity W_{ij} gives a simple attainability test. Even when $W_{ij} < L_c$, any feasible construction with $U_c = L_c$ establishes exactness. Otherwise $[L_c, U_c]$ is the reported identification interval for the optimal budget. For winner loss we compute every rival's lower bound and use

$$\min_{j \neq \ell} L_{\ell j} \;\leq K^*_{\text{win}} \;\leq \min_{j \neq \ell} U_{\ell j}. \tag{5}$$

Rivals whose lower bound exceeds an existing feasible upper bound need no further search. A jointly fitted home-effect covariate changes the inverse-Hessian structure; we use only feasible upper bounds for that control.

3.3 Constructions, noise, and uncertainty

Our static influence construction orders eligible records by positive estimated gap reduction,

$$g_r = (2y_r - 1) c^\top H(\hat{\theta})^{-1} x_r. \tag{6}$$

It reverses records in descending order, with stable row-order tie breaking. We refit *every integer prefix* until a strict crossing; U_c is the first crossing on this specific path. It is a global optimum only when a lower bound agrees. The local prediction is the first prefix whose cumulative g_r exceeds $\hat{\Delta}$. Controls recompute the influence order after each edit, restrict static selection to comparisons incident to either queried item, or delete records instead. For deletion, the ordering score is $(y_r - \sigma(x_r^\top \hat{\theta})) c^\top H^{-1} x_r$ and ties are eligible. Its result is an upper bound for a different threat model.

Uniform noise samples decisive records without replacement, independently of outcome or influence. Each of 500 seeded permutations is evaluated at fixed global fractions from 0.01% to 10%, with budget $k = \lceil fN \rceil$. The same permutation supplies nested budgets; different replicates are independent.

Table 1: Real-data main panels. n_ℓ is original leader support. Intervals use record/game resampling or question clusters, as applicable. NBA years denote season-ending years. Scores are our ridge BT reconstruction.

Panel	N	M	Tie (%)	n_ℓ	$\widehat{\Delta}$	Bootstrap 95% interval
Arena	57,477	64	30.9	7387	0.008	[-0.097, 0.116]
MT-human	3,355	6	23.2	1029	0.376	[0.124, 0.654]
MT-GPT4	2,400	6	25.0	800	0.809	[0.480, 1.090]
NBA-2010	1,230	30	0.0	82	0.117	[-0.503, 0.772]
NBA-2011	1,230	30	0.0	82	0.062	[-0.676, 0.722]
NBA-2012	990	30	0.0	66	0.106	[-0.663, 0.895]
NBA-2013	1,229	30	0.0	82	0.279	[-0.292, 0.963]
NBA-2014	1,230	30	0.0	82	0.185	[-0.474, 0.819]
NBA-2015	1,230	30	0.0	82	0.546	[-0.192, 1.203]

We record leading-pair reversal, any winner loss, and normalized Kendall discordance over all item pairs. Wilson 95% intervals describe Monte Carlo uncertainty conditional on the observed data, not uncertainty about a population corruption rate.

We separately resample the original observations 500 times: questions as clusters in MT-Bench, records in Arena, and games within a season in the NBA. Intervals are percentile intervals for the *originally selected, fixed* score contrast. They are descriptive, do not adjust for selecting the leading pair, and should not be treated as post-selection hypothesis tests. As a resolution check, let s be that bootstrap gap standard deviation. We also construct a reversal satisfying $c^\top \hat{\theta}_E < -2s$, yielding U_{2s}. This uses a frozen original uncertainty scale; it is not a confidence interval for the edited, adaptively selected dataset.

4 Data and experimental protocol

Public snapshots. Arena uses the supplied 57,477-record human-preference CSV, keeping all 64 model identifiers distinct. MT-human uses all 3,355 judgments, including expert and author annotators, and MT-GPT4 uses 2,400 pairwise judgments on the same six model names and 80 questions. Both ordinary ties and GPT-4's `tie (inconsistent)` outcomes receive $y = 1/2$. Human judgments include repeated annotators and turns; they are not independent votes on unrelated prompts. The NBA archive [FiveThirtyEight, 2025] is reduced to one row per game using `_iscopy=0`, NBA league only, regular seasons only, separately for 2010–2015. We use game winners, not the supplied Elo ratings or scores; the reconstructed score leader is not necessarily the playoff champion. The shortened 2012 season and the 1,229-game 2013 season are retained as recorded. The latter count is not padded to an expected schedule. Table 1 gives all panel sizes. No synthetic data enter the empirical findings; small synthetic fixtures only verify the mathematics and implementation.

Matched design control. After canonicalizing participant order and complementing outcomes when needed, we intersect MT-Bench keys (question, turn, unordered model pair). One human judgment is sampled per common key, while the GPT-4 record is unique. This yields identical schedules of 1,814 records, retaining question clusters. The main matched seed is 20261102; 100 human selections assess annotator-choice variability. The restriction equalizes item identities, prompts, turns, pair multiplicities, and ridge strength. It does not randomize judge type, align rubrics, or make the human panel independent of the original study's collection process.

Implementation and checks. We aggregate exact pair/outcome sufficient statistics, solve the convex objective with damped Newton steps, and reject fits with maximum gradient residual above 10^{-6}. Near ties use conservative numerical guards. All empirical thresholds are checked by refitting actual records; certificates are analytic bounds evaluated in floating point, not interval-arithmetic proofs. Five verification tests include comparison with an independent optimizer, finite-difference influence, orientation and replication invariance, and exhaustive small-instance checks of both bounds. The seed is 20261102. The main experiment took 95.0 s and the all-rival winner audit 11.0 s on an Intel Xeon Gold 6230 CPU with one BLAS thread; main-process peak resident memory was

Table 2: Certified optimal-count intervals and constructed budgets. A single number means lower and feasible upper bounds agree, hence exactness. U is the static leading-pair construction, even where an interval remains. Winner loss considers every rival and may precede runner-up reversal. U_{2s} crosses the frozen two-bootstrap-SD margin. Percentages use the pair construction U.

Panel	Pair K^*	Winner K^*_{win}	$100U/N$	$100U/n_\ell$	U_{2s}
Arena	2	2	0.0035	0.027	27
MT-human	46	46	1.371	4.470	78
MT-GPT4	61	61	2.542	7.625	93
NBA-2010	2	2	0.163	2.439	13
NBA-2011	1	1	0.081	1.220	11
NBA-2012	[1, 2]	[1, 2]	0.202	3.030	11
NBA-2013	3	3	0.244	3.659	14
NBA-2014	[2, 4]	[2, 4]	0.325	4.878	17
NBA-2015	[5, 8]	[5, 7]	0.650	9.756	21

344.7 MiB. Appendix C and the artifact specify versions, commands, sources, checksums, and per-display provenance.

5 Results

5.1 Exact fragility and unresolved order are different facts

Table 2 identifies the exact leading-pair and any-winner-loss minima in 6 of the nine main panels. Arena requires exactly 2 reversals, compared with exactly 46 for MT-human and 61 for MT-GPT4. These are certified minima over all eligible record sets, not only successes of the influence heuristic. The respective direct-win capacities are 31, 91, and 75, enough to attain the pair relaxation; all-rival bounds also rule out cheaper winner changes.

The implications differ sharply. Arena's original leading gap is only 0.008; the fixed pair reverses in 44.2% of the original-data bootstrap resamples. Its point ordering is unresolved at that sampling scale. Requiring a gap below minus twice the original standard deviation increases the constructed budget from 2 to 27. The analogous budgets are 78 and 93 for the separated MT-Bench leaders. These larger numbers are feasible construction counts, not claimed optimal minima or post-edit significance tests.

NBA panels require few games globally, but not uniformly the same number. The original leading-pair bounds range from a single outcome in 2011 to $[5, 8]$ in 2015. Their baseline bootstrap intervals all contain zero. The NBA-2015 winner-loss interval is $[5, 7]$: a constructed path promotes Houston before the original runner-up Atlanta overtakes Golden State. This illustrates why a leading-pair endpoint cannot stand in for all winner changes.

5.2 Random noise is much less effective than selected edits

At a global 1% noise budget, Arena's original leading pair reverses in 31.0% of runs (Wilson interval 27.1–35.2%). Neither full MT-Bench panel reverses in any of the 500 runs, whose Wilson upper limit is 0.8%. Even at 10% random reversals, neither full MT-Bench leading pair reverses in the observed runs. This is conditional simulation evidence, not proof that their reversal probability is zero. The NBA seasons vary substantially at the same fractions (Figure 1). The point-order threshold alone therefore does not describe a board's response to unselected label noise.

Small changes can also disturb less prominent ranks without removing the winner. At 1% random noise, mean normalized Kendall discordance in Arena is 0.0115 despite its much larger leading-pair reversal probability. Conversely, full MT-human has occasional lower-rank changes at that budget with no winner change. Appendix B reports both endpoints and empirical discordance ranges. These statistics measure distinct aspects of stability and should not be substituted for one another.

5.3 Exposure changes the cross-design comparison

On the global scale, the MT-GPT4 leading-pair minimum is 2.5417%, while every NBA constructed leading-pair budget is below 1% (Table 2). But GPT-4 contributes 800 of 2,400 MT-GPT4 records,

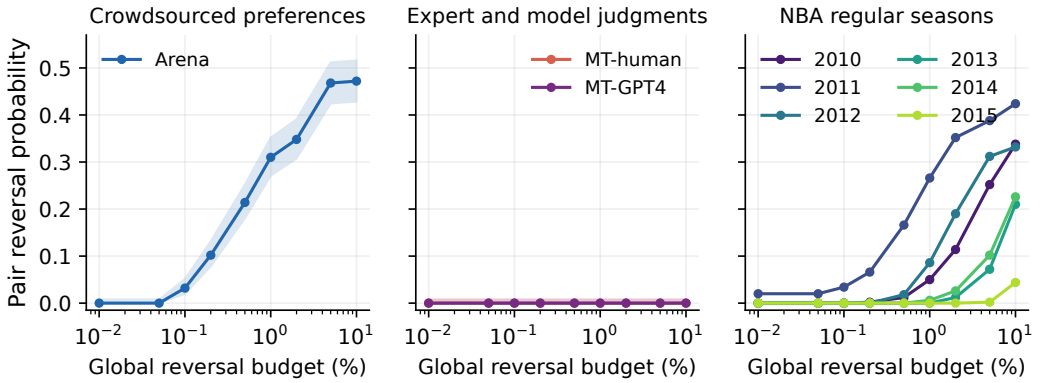


Figure 1: Leading-pair reversal under uniform decisive-outcome noise. Each point uses 500 independent permutations. Shading shows pointwise Wilson 95% Monte Carlo intervals for Arena and MT-Bench; NBA season curves are separate, without shaded intervals for readability. Small datasets round several small fractions to the same integer budget.

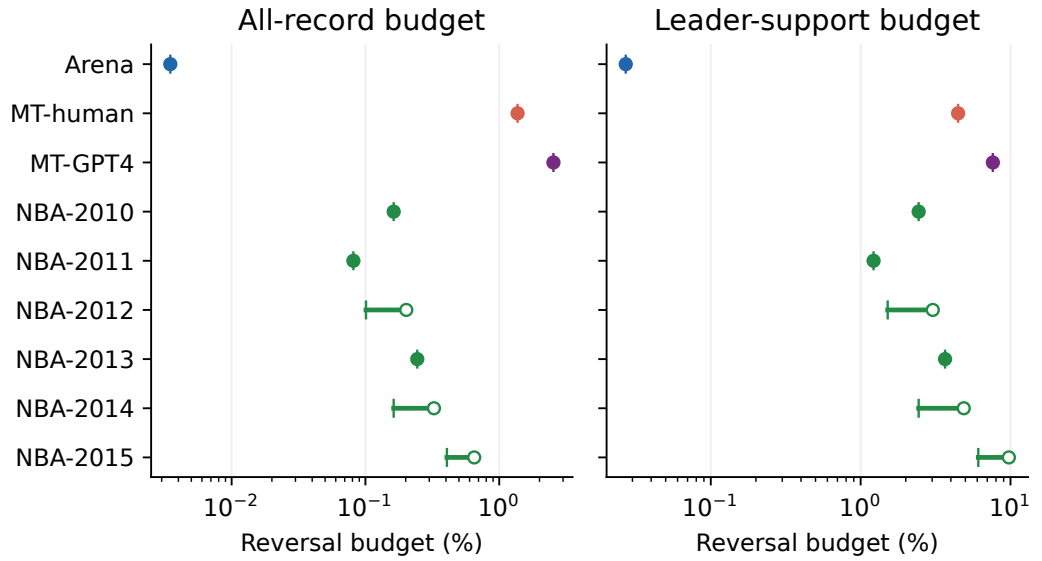


Figure 2: The same leading-pair count bounds under two denominators. Filled circles are exact minima; open circles are feasible upper bounds, with a segment to the certified lower bound. Leader support includes ties and remains fixed at the original dataset. The two scales describe exposure, not equal corruption effort.

compared with 82 of 1,230 games for the 2015 NBA leader. The same MT-GPT4 threshold is 7.62% of leader support; the 2015 NBA pair construction is 9.76%. The ordering of these *upper/construction percentages* changes, while their certified intervals on the leader scale overlap. We therefore cannot conclude that NBA-2015 is intrinsically more robust. We can conclude that a global percentage alone exaggerates how comparable the supporting evidence is across these boards.

Arena remains unusually close under either scale, but that is a statement about this snapshot and pair. Within its top ten, adjacent-pair constructions range from 2 to 104 edits. Across all panels, including matched controls, the lower bound and feasible construction agree on 72 of 83 queried adjacent pairs. A single winner threshold consequently omits substantial within-board heterogeneity.

Table 3: Identical MT-Bench comparison schedules, one human judgment per common key, main seed. Both leading-pair minima are exact. Bootstrap intervals resample questions.

Panel	N	$\widehat{\Delta}$	Bootstrap 95% interval	K^*	$100K^*/N$
Matched-human	1,814	0.283	[0.066, 0.496]	20	1.103
Matched-GPT4	1,814	0.759	[0.441, 1.107]	42	2.315

5.4 Matched judgments and modeling controls

Under the matched schedule, the human threshold is 20 and the GPT-4 threshold is 42 (Table 3); their gaps also differ. Over 100 selections of the human annotator per key, the human construction has median 20 and observed range 15–26. The matched GPT-4 dataset is unchanged across those selections. This controls several schedule differences, but does not prove a judge-design effect: the GPT-4 judge evaluates GPT-4's own responses, and greater score separation could reflect self-preference or a different rubric rather than greater accuracy [Zheng et al., 2023].

Recomputed and static influence have the same leading-pair counts in every studied panel; restricting candidates to incident comparisons also leaves those counts unchanged. This is partly expected when the direct-win relaxation is attainable. It does not imply that remote comparisons never matter in other schedules. First-order counts match the exact-refit prefix in 65 of 83 adjacent-pair queries, with mean absolute count error 0.73 and maximum 9. Approximation accuracy is useful for finding edits, but is not an optimality certificate.

The modeling controls materially qualify the strongest-looking result. Arena's threshold becomes 1 at $\lambda = 0.1$ and 11 at $\lambda = 10$; dropping ties reverses the *original* leading pair before any edit. The full MT-Bench thresholds remain 46 and 61 under these ridge changes, but dropping ties reduces them to 42 and 40, respectively, on the smaller decisive-only datasets. These are distinct estimands, not competing ways to estimate one immutable number. Adding a fitted home effect to each NBA season leaves the reported leading-pair construction counts unchanged; the theorem is not applied to those covariate fits. Deletion needs 3, 92, and 144 records for Arena, MT-human, and MT-GPT4 under its own static path. Appendix B reports all controls, including the weaker curvature-only lower bound.

6 Discussion, limitations, and responsible use

The defensible conclusion is conditional: in these reconstructed fixed-schedule boards, targeted edits can be much more effective than random noise, but a tiny reversal can concern an already unresolved order. We recommend reporting a certified interval or a clearly labeled feasible budget, the original gap with an uncertainty scale, and both global and participant-support fractions. A robust leading ordering can coexist with unstable lower ranks. A stable model-judge board can encode a systematic evaluation preference.

The finite-edit result is specific. It assumes standard item-only ridge BT, positive diagonal regularization, fixed comparison weights, and unit reversals of decisive labels. It does not cover arbitrary judge or home covariates, adaptive scheduling, new comparisons, removal of records, tie corruption, or corruption restricted by user identity. After direct-win capacity is exhausted the relaxation can be unattainable, as in several NBA seasons; the remaining count intervals are not silently resolved by a heuristic. The conservative analytic proofs are not machine-verified floating-point certificates.

The empirical comparison is descriptive rather than causal. There is one historical Arena snapshot and one small MT-Bench question collection; the NBA seasons are related cohorts in one sport. Model identities, sample sizes, margins, schedules, and stakes differ. Question clustering handles within-question dependence but not all shared-annotator effects; Arena lacks usable user clusters here, and game resampling ignores temporal, roster, and schedule dependence. Selection of top pairs also makes the bootstrap intervals descriptive. No interval estimates a universal population adversarial budget. We did not measure calibration, factual correctness, fairness, or the validity of the underlying judgments, nor run live attacks or evaluate defenses.

The audit can support more cautious leaderboard communication, but the edit-selection code could also inform manipulation. Its scope is offline analysis of public historical outcomes. We made no platform submissions or participant contacts, and release no conversation text or user profiles. The

intended use is to audit uncertainty and evidence concentration, with qualified count bounds rather than assertions about deployed-system security. An OpenAI Codex assistant developed the methods, wrote and ran the code, inspected the results, and drafted the manuscript. No paid model calls generated judgments or experimental measurements; all reported evidence comes from the supplied public-data snapshots and actual CPU runs. This disclosure is essential to interpreting the artifact's provenance, and is not a substitute for independent scientific review.

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A Proofs and scope of the certificates

A.1 Maximum principle and direct-transfer domination

Lemma 1 (Endpoint extrema)**.** *For $H = L_w + \lambda I$ with L_w a symmetric graph Laplacian, nonnegative weights, and $\lambda > 0$, let $v = H^{-1}(e_i - e_j)$ for $i \neq j$. Then $v_i = \max_m v_m > 0$ and $v_j = \min_m v_m < 0$.*

Proof. Since $H\mathbf{1} = \lambda\mathbf{1}$, we have $\sum_m v_m = 0$. Also $v \neq 0$, so its maximum is positive and minimum is negative. If a positive maximum occurred at a node $m \neq i$, then

$$(Hv)_m = \lambda v_m + \sum_q w_{mq}(v_m - v_q) > 0. \tag{7}$$

This contradicts $(e_i - e_j)_m \leq 0$. Thus the maximum is attained only at i. The corresponding argument at a negative minimum shows it is attained only at j. Connectivity is not needed because ridge is positive at every node. □

Proof of Theorem 1. Equation (1) has Hessian

$$H(\theta) = \sum_r \sigma(x_r^\top \theta)[1 - \sigma(x_r^\top \theta)] x_r x_r^\top + \lambda I. \tag{8}$$

It is a positive definite weighted Laplacian plus ridge, independent of outcomes at a fixed θ. Its smallest eigenvalue is at least λ, so adding any linear term preserves coercivity, strong convexity, and a unique smooth optimizer.

Write $b_E = \sum_{q=1}^k u_q$, where every $u_q = e_{w_q} - e_{l_q}$. For $s \in [0, 1]$, let $b(s) = (1 - s)b_E + skc$ and define $\theta(s)$ by $\nabla F(\theta(s)) + b(s) = 0$. Implicit differentiation yields

$$\frac{d}{ds} c^\top \theta(s) = -v(s)^\top (kc - b_E), \qquad v(s) = H(\theta(s))^{-1} c. \tag{9}$$

By Lemma 1, $v(s)^\top u_q \leq v_i(s) - v_j(s) = v(s)^\top c$. Therefore the derivative in Eq. (9) is nonpositive. Integrating from zero to one proves Eq. (4). This step is why the argument is finite-edit rather than a first-order extrapolation at $\hat{\theta}$.

Along tc, implicit differentiation gives $d(c^\top \theta_c(t))/dt = -c^\top H(\theta_c(t))^{-1} c < 0$. The integer crossing in Eq. (3) exists: the non-ridge gradient is bounded for a fixed finite schedule, whereas the ridge estimating equation implies $\theta_c(t) = -tc/\lambda + O(1)$ as $t \to \infty$. If $k < L_c$, the virtual gap is nonnegative, hence so is the gap for every feasible k-edit set. If $W_{ij} \geq L_c$, reversing L_c distinct original i-over-j wins yields exactly the linear term $L_c c$ and crosses. This proves both the bound and attainability. □

This theorem concerns the fixed queried contrast, not all ranks simultaneously. For original-winner loss, a strict reversal with any rival suffices, so taking the minimum over all rivals establishes Eq. (5). Upper/lower equality can also establish exactness without direct-capacity attainment. For example, NBA-2013 has direct capacity 2, pair lower bound 3, and a feasible three-edit construction. Bounds across rivals are recomputed from the original data, not from an already edited board.

A.2 A generic curvature bound used as a comparison

We also implemented a less specialized finite-edit bound to quantify the benefit of exploiting the item-comparison structure. Let $H_0 = H(\hat{\theta})$, $a = \max_r \|x_r\|_{H_0^{-1}}$, and $\rho = \hat{\Delta}/\|c\|_{H_0^{-1}}$. Sort the norms of eligible records in descending order and let S_k be the sum of the largest k. Then

$$S_k < \frac{1 - e^{-a\rho}}{a} \quad \Longrightarrow \quad c^\top \hat{\theta}_E > 0 \text{ for every } |E| \leq k. \tag{10}$$

To prove it, the logistic curvature $w(z) = \sigma(z)(1 - \sigma(z))$ obeys $|d \log w(z)/dz| \leq 1$. For $d = \theta - \hat{\theta}$ and $R = \|d\|_{H_0}$, Cauchy–Schwarz gives $|x_r^\top d| \leq aR$, hence $H(\hat{\theta} + ud) \succeq e^{-uaR} H_0$ for $u \in [0, 1]$

(the ridge term also satisfies this inequality). At the edited optimum, with $q = \|b_E\|_{H_0^{-1}}$,

$$Rq \;\geq -d^\top b_E = \int_0^1 d^\top H(\hat{\theta} + ud)\, d\, du \geq \frac{R}{a}(1 - e^{-aR}). \tag{11}$$

For $R > 0$, Eq. (11) and $q \leq S_k$ imply $R < \rho$ under Eq. (10). Such a displacement cannot reach the contrast-zero hyperplane. The $R = 0$ case is immediate. Code adds the inverse-Hessian norm of any initial numerical gradient residual to S_k and a strict numerical guard. Table 4 reports this certificate as a lower bound $L_{\mathrm{curv}} = k_{\mathrm{safe}} + 1$ on the minimum count, not as an attack. Failure to certify even one edit is not evidence that one edit suffices.

B Supplementary measurements

Table 4: Constructed leading-pair thresholds and lower bounds. $W_{\ell r}$ is direct-win capacity; L_{dir} is Theorem 1's necessary count; L_{curv} is the curvature-only necessary count. Adaptive and incident columns are feasible flip constructions. Delete uses its own operation and ordering and has no direct-transfer certificate.

Panel	$W_{\ell r}$	L_{dir}	U	L_{curv}	Adaptive	Incident	Delete
Arena	31	2	2	1	2	2	3
MT-human	91	46	46	31	46	46	92
MT-GPT4	75	61	61	27	61	61	144
NBA-2010	2	2	2	1	2	2	3
NBA-2011	1	1	1	1	1	1	2
NBA-2012	0	1	2	1	2	2	2
NBA-2013	2	3	3	2	3	3	5
NBA-2014	0	2	4	2	4	4	4
NBA-2015	1	5	8	3	8	8	9
Matched-human	51	20	20	14	20	20	39
Matched-GPT4	55	42	42	20	42	42	96

Table 5: Uniform noise at 1% of all records. Pair and winner columns are event percentages across 500 permutations. Wilson intervals measure Monte Carlo error for the pair probability. D_K is discordant item pairs divided by $\binom{M}{2}$; its central range is empirical across perturbations, not a confidence interval for its mean.

Panel	Pair (%)	Wilson 95% interval	Winner (%)	Mean D_K	Central 95% D_K
Arena	31.0	[27.1, 35.2]	31.0	0.0115	[0.0069, 0.0174]
MT-human	0.0	[0.0, 0.8]	0.0	0.0005	[0.0000, 0.0000]
MT-GPT4	0.0	[0.0, 0.8]	0.0	0.0000	[0.0000, 0.0000]
NBA-2010	5.0	[3.4, 7.3]	7.8	0.0222	[0.0092, 0.0345]
NBA-2011	26.6	[22.9, 30.6]	26.6	0.0174	[0.0069, 0.0288]
NBA-2012	8.6	[6.4, 11.4]	9.0	0.0234	[0.0092, 0.0368]
NBA-2013	0.0	[0.0, 0.8]	0.0	0.0185	[0.0092, 0.0299]
NBA-2014	0.6	[0.2, 1.7]	0.6	0.0147	[0.0046, 0.0253]
NBA-2015	0.0	[0.0, 0.8]	0.0	0.0201	[0.0092, 0.0322]
Matched-human	0.0	[0.0, 0.8]	0.0	0.0000	[0.0000, 0.0000]
Matched-GPT4	0.0	[0.0, 0.8]	0.0	0.0000	[0.0000, 0.0000]

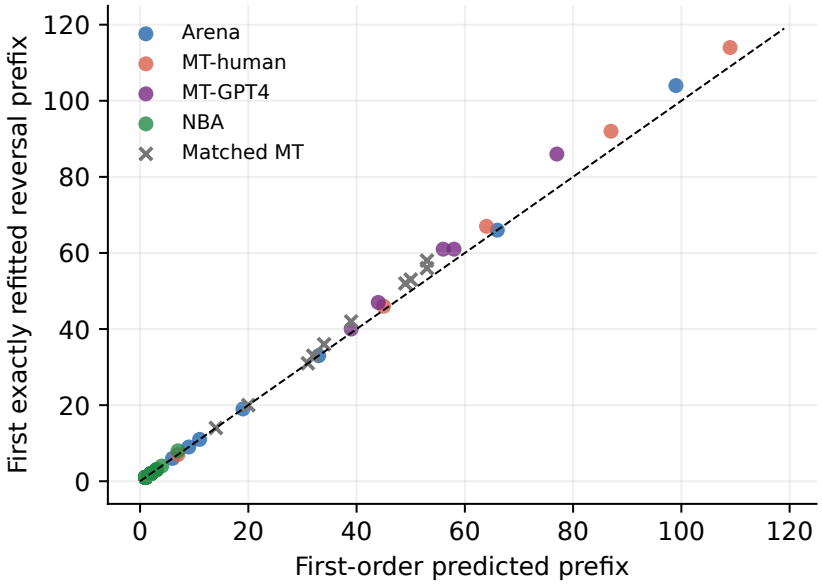


Figure 3: Initial influence prediction versus exactly refitted first crossing along its sorted path. These 83 queries cover all adjacent pairs among the original top ten (or all six models for MT-Bench), including matched controls. Coincident markers are not jittered. Prediction agreement does not establish global optimality.

Table 6: All modeling controls. The original main-fit pair is held fixed, so a zero threshold means the alternative modeling choice already reverses it. Ridge is on the summed loss. Drop-ties changes the observation count and loss weights; these counts are not percentages on the original denominator. Home-covariate fits are upper-bound constructions only.

Panel	Variant	Fitted winner	Original-pair gap	U
Arena	ridge-0.1	gpt-4-1106-preview	0.0042	1
Arena	ridge-10	gpt-4-1106-preview	0.0436	11
Arena	drop-ties	gpt-4-0125-preview	-0.0226	0
MT-human	ridge-0.1	gpt-4	0.3781	46
MT-human	ridge-10	gpt-4	0.3574	46
MT-human	drop-ties	gpt-4	0.5270	42
MT-GPT4	ridge-0.1	gpt-4	0.8198	61
MT-GPT4	ridge-10	gpt-4	0.7245	61
MT-GPT4	drop-ties	gpt-4	1.0645	40
NBA-2010	ridge-0.1	CLE	0.1272	2
NBA-2010	ridge-10	CLE	0.0697	2
NBA-2010	home-covariate	CLE	0.1272	2
NBA-2011	ridge-0.1	SAS	0.0729	1
NBA-2011	ridge-10	SAS	0.0170	1
NBA-2011	home-covariate	SAS	0.0668	1
NBA-2012	ridge-0.1	SAS	0.1278	2
NBA-2012	ridge-10	SAS	0.0347	1
NBA-2012	home-covariate	SAS	0.1037	2
NBA-2013	ridge-0.1	MIA	0.3044	3
NBA-2013	ridge-10	MIA	0.1740	3
NBA-2013	home-covariate	MIA	0.2909	3
NBA-2014	ridge-0.1	SAS	0.2016	4
NBA-2014	ridge-10	SAS	0.1090	4
NBA-2014	home-covariate	SAS	0.1946	4
NBA-2015	ridge-0.1	GSW	0.6122	8
NBA-2015	ridge-10	GSW	0.2865	7
NBA-2015	home-covariate	GSW	0.5579	8
Matched-human	ridge-0.1	gpt-4	0.2865	20
Matched-human	ridge-10	gpt-4	0.2561	20
Matched-human	drop-ties	gpt-4	0.4278	19
Matched-GPT4	ridge-0.1	gpt-4	0.7722	42
Matched-GPT4	ridge-10	gpt-4	0.6550	42
Matched-GPT4	drop-ties	gpt-4	0.9882	28

B.1 All adjacent-pair queries

The following tables give the fitted pair, baseline gap, certified optimal-count interval, and first-order prefix prediction for every query. Rows are selected by the original ranking and are descriptive, without multiplicity-adjusted inferential claims. Original model identifiers are preserved.

Table 7: Adjacent pairs: Arena.

Position	Higher	Lower	Gap	K^*	Pred.
1	gpt-4-1106-preview	gpt-4-0125-preview	0.008	2	2
2	gpt-4-0125-preview	gpt-4-0314	0.405	[82, 104]	99
3	gpt-4-0314	gpt-4-0613	0.108	66	66
4	gpt-4-0613	qwen1.5-72b-chat	0.074	9	9
5	qwen1.5-72b-chat	mistral-medium	0.010	2	2
6	mistral-medium	claude-1	0.040	[17, 19]	19
7	claude-1	claude-2.0	0.086	33	33
8	claude-2.0	gemini-pro-dev-api	0.041	[9, 11]	11
9	gemini-pro-dev-api	gemini-pro	0.016	[3, 6]	6

Table 8: Adjacent pairs: MT-human.

Position	Higher	Lower	Gap	K^*	Pred.
1	gpt-4	claude-v1	0.376	46	45
2	claude-v1	gpt-3.5-turbo	0.042	7	7
3	gpt-3.5-turbo	vicuna-13b-v1.2	0.654	114	109
4	vicuna-13b-v1.2	alpaca-13b	0.701	92	87
5	alpaca-13b	llama-13b	0.678	67	64

Table 9: Adjacent pairs: MT-GPT4.

Position	Higher	Lower	Gap	K^*	Pred.
1	gpt-4	claude-v1	0.809	61	56
2	claude-v1	gpt-3.5-turbo	0.458	[39, 40]	39
3	gpt-3.5-turbo	vicuna-13b-v1.2	0.688	61	58
4	vicuna-13b-v1.2	alpaca-13b	1.044	86	77
5	alpaca-13b	llama-13b	0.684	47	44

Table 10: Adjacent pairs: NBA-2010.

Position	Higher	Lower	Gap	K^*	Pred.
1	CLE	ORL	0.117	2	2
2	ORL	LAL	0.056	1	1
3	LAL	DAL	0.134	2	2
4	DAL	PHO	0.061	1	1
5	PHO	UTA	0.019	1	1
6	UTA	DEN	0.003	1	1
7	DEN	ATL	0.046	1	1
8	ATL	SAS	0.105	1	1
9	SAS	OKC	0.006	1	1

Table 11: Adjacent pairs: NBA-2011.

Position	Higher	Lower	Gap	K^*	Pred.
1	SAS	CHI	0.062	1	1
2	CHI	DAL	0.169	2	2
3	DAL	LAL	0.026	1	1
4	LAL	MIA	0.029	1	1
5	MIA	OKC	0.084	1	1
6	OKC	BOS	0.018	1	1
7	BOS	ORL	0.209	2	2
8	ORL	DEN	0.014	1	1
9	DEN	POR	0.077	1	1

Table 12: Adjacent pairs: NBA-2012.

Position	Higher	Lower	Gap	K^*	Pred.
1	SAS	CHI	0.106	[1, 2]	2
2	CHI	OKC	0.078	1	1
3	OKC	MIA	0.144	2	2
4	MIA	LAL	0.248	[2, 3]	3
5	LAL	MEM	0.009	1	1
6	MEM	LAC	0.083	1	1
7	LAC	IND	0.006	1	1
8	IND	DEN	0.105	[1, 2]	2
9	DEN	ATL	0.015	1	1

Table 13: Adjacent pairs: NBA-2013.

Position	Higher	Lower	Gap	K^*	Pred.
1	MIA	OKC	0.279	3	3
2	OKC	SAS	0.089	1	1
3	SAS	DEN	0.058	1	1
4	DEN	MEM	0.073	1	1
5	MEM	LAC	0.012	1	1
6	LAC	NYK	0.175	2	2
7	NYK	IND	0.257	3	3
8	IND	GSW	0.010	1	1
9	GSW	BRK	0.011	1	1

Table 14: Adjacent pairs: NBA-2014.

Position	Higher	Lower	Gap	K^*	Pred.
1	SAS	OKC	0.185	[2, 4]	4
2	OKC	LAC	0.121	2	2
3	LAC	HOU	0.140	2	2
4	HOU	POR	0.014	1	1
5	POR	IND	0.053	1	1
6	IND	MIA	0.082	1	1
7	MIA	GSW	0.023	1	1
8	GSW	MEM	0.038	1	1
9	MEM	DAL	0.064	1	1

Table 15: Adjacent pairs: NBA-2015.

Position	Higher	Lower	Gap	K^*	Pred.
1	GSW	ATL	0.546	[5, 8]	7
2	ATL	HOU	0.095	1	1
3	HOU	LAC	0.007	1	1
4	LAC	MEM	0.055	1	1
5	MEM	SAS	0.010	1	1
6	SAS	CLE	0.190	[2, 3]	3
7	CLE	POR	0.036	1	1
8	POR	DAL	0.012	1	1
9	DAL	CHI	0.129	2	2

Table 16: Adjacent pairs: Matched-human.

Position	Higher	Lower	Gap	K^*	Pred.
1	gpt-4	claude-v1	0.283	20	20
2	claude-v1	gpt-3.5-turbo	0.175	14	14
3	gpt-3.5-turbo	vicuna-13b-v1.2	0.672	56	53
4	vicuna-13b-v1.2	alpaca-13b	0.725	52	49
5	alpaca-13b	llama-13b	0.526	31	31

Table 17: Adjacent pairs: Matched-GPT4.

Position	Higher	Lower	Gap	K^*	Pred.
1	gpt-4	claude-v1	0.759	42	39
2	claude-v1	gpt-3.5-turbo	0.485	33	32
3	gpt-3.5-turbo	vicuna-13b-v1.2	0.726	53	50
4	vicuna-13b-v1.2	alpaca-13b	0.945	58	53
5	alpaca-13b	llama-13b	0.742	36	34

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